Statistical Analysis of MLB Players: Performance, Experience, and Salary

Dara Friel 40403217, Katie-Marie Grant 40357584, Benjamin Hamilton 40405601, Eimear Hegarty 40362924, and Ryan Whitten 40368855

1 Abstract

This report analyses a dataset containing performance statistics, salary information, and team associations for 322 Major League Baseball (MLB) players from the 1986 season and their respective salaries for 1987. The dataset includes offensive and defensive variables, career totals up to that point, and positional data, allowing for a comprehensive analysis of player performance and impact on salary.

To evaluate the relationship between offensive and defensive skills, we conducted a Spearman's correlation test, which showed little evidence of a strong connection between hitting and defensive efficiency. Further analysis identified standout players by assessing their offensive, defensive, and overall performance, highlighting the player who excelled in each category.

In our investigation we also examined league differences, showing us that the teams in the American League (AL) tended to have higher-paid players compared to those in the National League (NL). We also found that a player's experience has a significant influence on their salary, with more experienced players generally earning higher wages. However, experience did not consistently correlate with improved performance data.

These findings provide meaningful insights into the factors that influence MLB salaries and enhance our understanding of how experience, performance, and league affiliation impact a player's success.

Contents

1 Abstract **1**

2 Outline of Dataset **3**
2.1 Introduction . . . 3
2.2 Dataset Overview . . . 3

3 Research Questions **5**

4 Results **5**
4.1 Initial Analysis . . . 5
4.1.1 Formatting of the Data . . . 5
4.1.2 Blunder Removal . . . 5
4.1.3 Test for Normality . . . 6
4.2 Is There Correlation Between Hitting and Defending? . . . 7
4.3 Do More Experienced Players Perform Better Offensively? . . . 10
4.4 Do More Experienced Players Perform Better Defensively? . . . 12
4.5 Does salary vary depending on team and league? . . . 13
4.6 Do more experienced players earn a higher salary? . . . 14
4.7 Modeling salaries of Major league Baseball Players . . . 15

5 Conclusions **21**
5.1 No Meaningful Correlation Between Hitting and Defence . . . 21
5.2 Experience Impacts Hitting Efficiency—But Only Slightly . . . 21
5.3 No Significant Difference in Defensive Efficiency by Experience . . . 21
5.4 Salary Varies Significantly by League and Team . . . 22
5.5 More Experienced Players Generally Earn Higher Salaries . . . 22
5.6 Modelling Salary of Major League Baseball players . . . 22

6 Further Analyses and Additional Data Considerations **23**
6.1 Player Consistency vs Variability . . . 23
6.2 Injury and Fatigue Factors . . . 23
6.3 Player Demographics . . . 23

7 Appendix **25**

2 Outline of Dataset

2.1 Introduction

Baseball is a sport deeply rooted in data analysis, where performance metrics are closely examined to evaluate player contributions and predict future success. Performance statistics are used to influence various aspects of the sport, from fan discussions to team managers' decisions and game strategies.
This dataset contains information about the 1986 Major League Baseball (MLB) season, containing a collection of the players' statistics from that year.
The dataset includes professional baseball players' offensive and defensive performance metrics from the 1986 season, their cumulative career statistics up to that point, and their salary information for the 1987 season.
Analysing these elements enables us to find meaningful and interesting insights into the potential relationships between on-field performance, salary compensation, and career progress.

2.2 Dataset Overview

The dataset contains the statistics of 322 baseball players across 23 different variables. Each row of the dataset represents a single hitter, with variables covering their statistics, performance, career history, team affiliations, and financial earnings. The variables can be organised into multiple categories:

Offensive Statistics
Offensive statistics reference the players who are batting and their aim is to reach home plate safely and score.

- Number of Times at Bat - the number of times a player has taken the plate to face the pitcher as a batter, excluding walks and sacrifices.
- Number of Hits - the number of successful times at bat i.e. times where the player has reached a base safely, not including walks.
- Number of Home runs - the number of home runs from successful hits.
- Number of Runs - the number of times a player has crossed home plate and scored.
- Number of Runs Batted In - the number of runs a player has contributed to by making it possible for their teammate to score.
  [Burns]
- Number of Walks - a walk occurs when a pitcher throws four pitches outside the strike zone that the batter does not attempt to hit, resulting in the batter being awarded first base.
  [MLB]

The dataset also records career totals in addition to the single-season statistics of 1986. This allows for analysis of long-term trends in the data, such as how a player's performance changes as they gain more experience and if there is a decline in the performance of veterans, who have been in the MLB for many years.

Defensive Statistics
Defensive statistics refer to the players who are out on the field, and their goal is to hinder the other team from scoring.

- Number of Putouts - a putout is awarded when a player is responsible for a play that results in an out. This is recorded when a player tags a runner out or catches a ball in the air.
- Number of Assists - an assist is recorded when a player aids another player in making a putout.
- Number of Errors - an error is marked against a player who makes a mistake in the field that allows a player on the opposing team to advance.
  [Butler]

The inclusion of all these statistics lets us run comparisons on defensive versus offensive statistics, and whether players excel more in one area than the other.

Experience and League Information

- Number of years in the Major Leagues.
- Player's League - There are two leagues in the MLB, the American League (AL) and the National League (NL).
- Player's Team - In 1986, there were 26 teams in the MLB, 14 in the AL, and 12 in the NL.
- Player's Position - Our dataset includes 25 different positions, the MLB has 10 traditional codes for positions and the others reference less common codes to reference the position.

This data is particularly valuable as it allows for an exploration of differences between the two leagues and provides insight into how a player's experience may influence their performance. The dataset contains information about the players' leagues and teams in both 1986 and 1987.

Salary Information

- Annual Salary on opening day in thousands of dollars, 1987.

One of the most valuable aspects of this dataset is the inclusion of 1987 salary information. This allows for insight into the correlation between salary and different variables, for example, how a player's experience translates into financial value.

This dataset offers a comprehensive view of professional baseball hitters, combining performance data with salary figures to enable meaningful statistical analysis. By examining the dataset through various techniques, we can identify patterns and relationships between players performance, experience and salary. The following section will present the specific questions and hypotheses our team has proposed to explore in greater detail.

3 Research Questions

We wished to investigate the correlation between hitting and defending performances, overall performance of players and if more experienced players perform better offensively or defensively. To do so we implemented Spearman's Correlation Test and Kruskal-Wallis test. When investigating salary, we tested it against team and league differences and the experience of individual players. To do so we used the Kruskal-Wallis test.

4 Results

4.1 Initial Analysis

4.1.1 Formatting of the Data

When inspecting the format of the data, we found that the salary was stored as a character variable, and therefore could not be used in calculations. We decided to change this to a numeric variable, so we could use it for further analysis. The code used to do this is as follows:

```
Data SOR2002.ProjectF_Cleaned;
    set SOR2002.Project_F_No_NA;
    Salary_int = input(salary , BEST12. );
    drop salary;
    rename Salary_int = salary;
run;
```

We also found that there was a spelling mistake in one of the variable labels. One of the variables was named *"no fo runs batted 86"*, which we corrected to *"no of runs batted 86"* using the following code:

```
proc datasets library=SOR2002 nolist;
   modify Project_F_No_NA;
   label 'no of runs batted 86'n = 'no of runs batted 86';
quit;
```

4.1.2 Blunder Removal

Upon a visual inspection of the data, we noticed a record where the values of all of the variables were 0. The name attached to the record was 'Mike Schmidt'. We decided to compare the record with another external, valid database and found that the results were a blunder. The record was removed from the dataset for all future proceedings using the following code:

```
data SOR2002.ProjectF_Cleaned;
    set SOR2002.ProjectF_Cleaned;
    if Name = "Mike Schmidt" Then Delete;
run;
```

4.1.3 Test for Normality

We performed normality tests on the variables under the following hypotheses:

- H_0: Data comes from a normal distribution.
- H_1: Data does not come from a normal distribution.

For the variable *"No. of times at bat 86"*, all p-values were less than 0.05, indicating statistical significance at the 5% level. Thus, we reject the null hypothesis and conclude that the data does not follow a normal distribution.

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.970192	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.069993	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.478686	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	3.059776	Pr > A-Sq	<0.0050

Figure 1: no of times at bat normality test

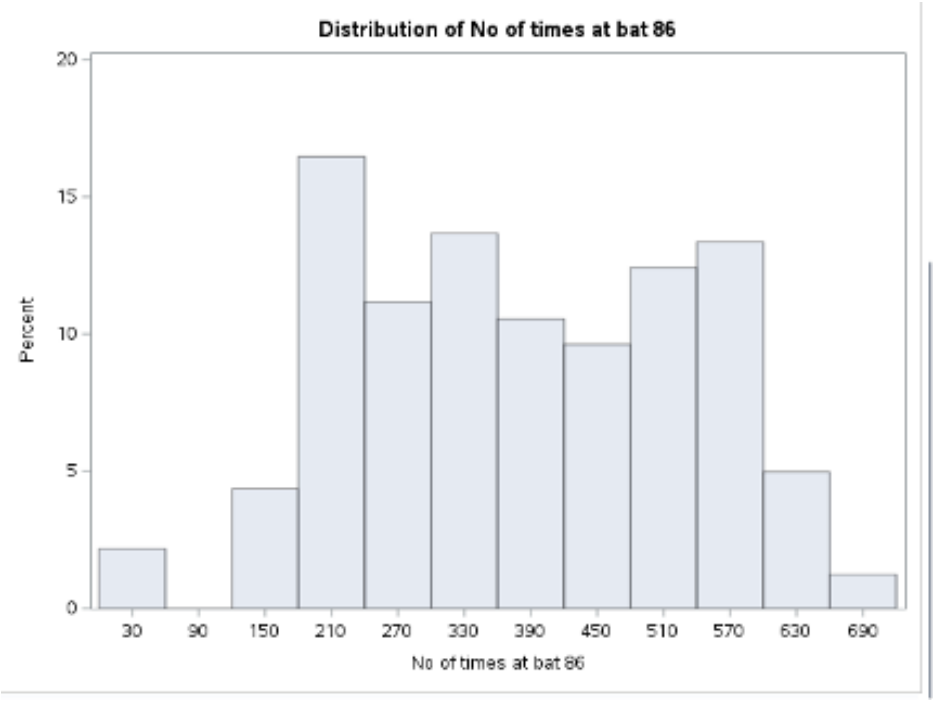


Figure 2: Distribution of No. of Times at Bat 86

A similar result was obtained for *"No. of home runs 86"*, confirming non-normality:

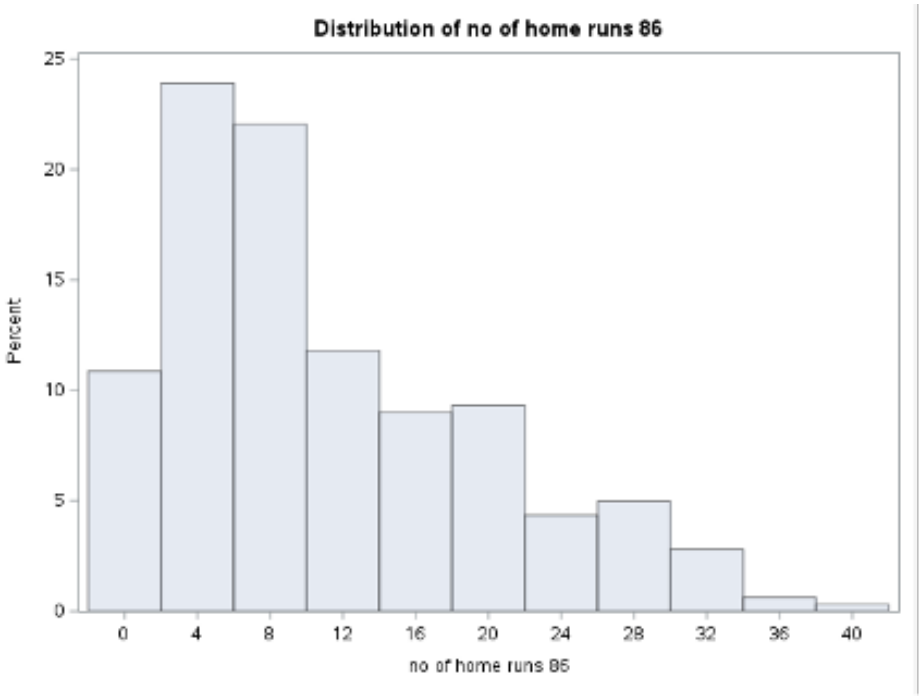


Figure 3: Distribution of No. of Home Runs 86

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.912484	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.148857	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.495479	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	8.85894	Pr > A-Sq	<0.0050

Figure 4: no of home runs normality test

This analysis was repeated for all variables, yielding similar results, suggesting that none follow a normal distribution, therefore all of the tests which we used from now on will be non-parametric as we cannot assume that the variables are normally distributed.

4.2 Is There Correlation Between Hitting and Defending?

Hitting efficiency metric

To calculate a players hitting performance, we introduce a new variable **Hitting Efficiency**. This variable incorporates home runs (HR), runs, runs batted in (RBI), hits and walks. This is weighted as follows:

$$\text{Hitting Efficiency} = \frac{(0.4 \times \text{HR} + 0.25 \times \text{Runs} + 0.2 \times \text{RBI} + 0.1 \times \text{Hits} + 0.05 \times \text{Walks})}{\text{Number of times at the bat}}$$

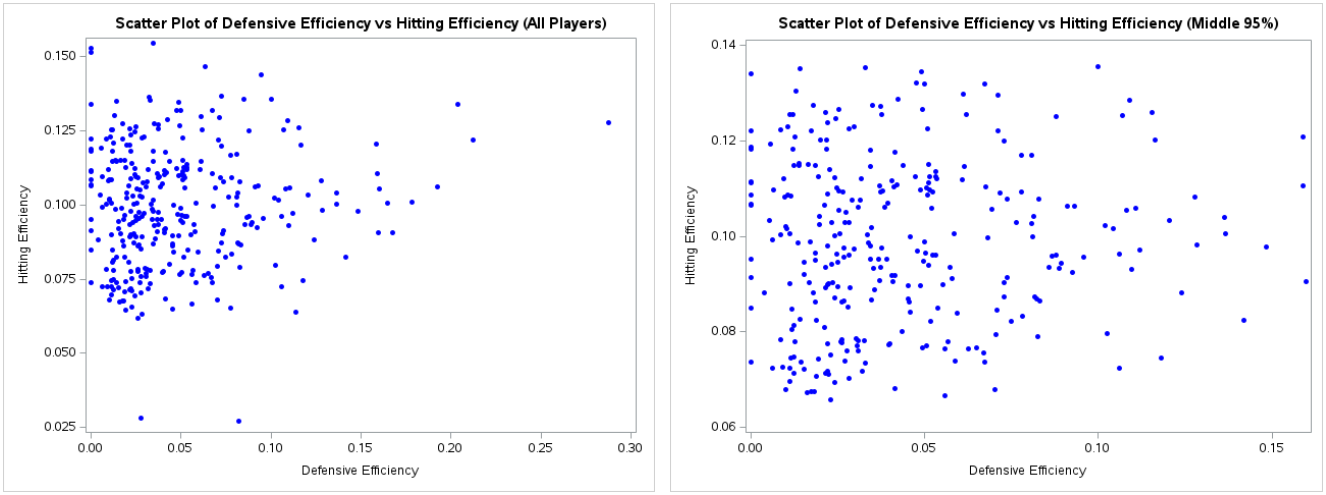
This metric gives higher weight to impactful plays (home runs) while incorporating hits and walks, providing a well-rounded measure of offensive efficiency.

Defensive efficiency metric
To evaluate a player's defensive performance, we introduce a new variable: **Defensive Efficiency**. This metric incorporates putouts, assists, and errors to provide a weighted measure of a player's fielding performance. This is seen by the following:

$$\text{Defensive Efficiency} = \frac{(0.001 \times \text{Putouts} + 0.0005 \times \text{Assists})}{(1 + \text{Errors})}$$

This metric rewards effective fielding while penalizing errors, ensuring a balanced evaluation of defensive performance.

To analyse the relationship between hitting and defensive efficiency, we plot the hitting efficiency of each player against their defensive efficiency. The first plot includes all players from the 1986 season, while the second plot displays only the middle 95% of data points around the median, filtering out potential outliers.



(a) All Players

(b) Filtered (Middle 95%)

Figure 5: Scatter plots of hitting efficiency vs. defensive efficiency. (a) Shows all players, while (b) filters out outliers, retaining only the middle 95% of data.

We can clearly see from (b) that there is no correlation between offensive and defensive efficiency. This can be confirmed by conducting a Spearman's Correlation Test.

Spearman's Correlation Test for Offensive and Defensive Efficiency
To assess the relationship between offensive efficiency and defensive efficiency, we conducted a non-parametric Spearman's correlation test at a 5% significance level ($\alpha = 0.05$).
Hypotheses:

- **Null Hypothesis (H_0)**: There is no correlation between offensive efficiency and defensive efficiency ($\rho_s = 0$).
- **Alternative Hypothesis (H_1)**: There is a significant correlation ($\rho_s \neq 0$).

Test Results:

- **Spearman's correlation coefficient** (ρ_s): 0.05991
- **p-value**: 0.2846

Since the p-value (0.2846) is greater than the significance level ($\alpha = 0.05$), we **fail to reject the null hypothesis**. This suggests that there is no statistically significant correlation between offensive efficiency and defensive efficiency based on Spearman's correlation.

Who is the Best Player?
Although the data does not show a strong correlation between hitting and defensive efficiency, it can still provide useful insights into some key questions:

1. Who was the best hitter in the 1986 season?
2. Who was the best defender in the 1986 season?
3. Who was the best overall player in the 1986 season?

To answer these questions, we analyse the top 100 players based on the number of times at bat (The most frequent batters). This selection ensures that we focus on players who have played consistently throughout the season, filtering out any outliers or "flash in the pan" performances. The following scatter plot visualizes these players:

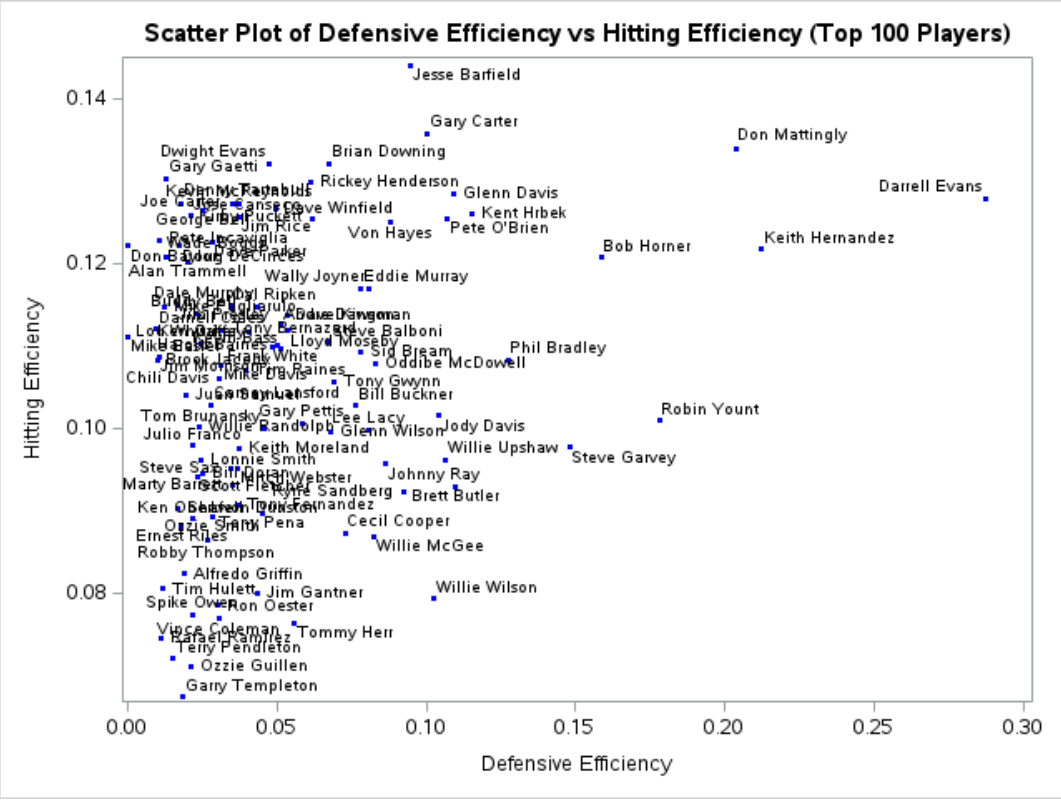


Figure 6: Scatter plot of hitting efficiency vs. defensive efficiency for the top 100 most frequent batters in the 1986 season.

Reducing the sample size allows us to label individual players on the graph, making it easier to identify standout performers. To do this we can use the following code:

```
proc sgplot data=top_at_bat_players;
    scatter x=defensive_efficiency y=weighted_hitting_efficiency /
            datalabel=player_name
            markerattrs=(symbol=circlefilled size=3 color=blue);

    xaxis label="Defensive Efficiency" min=0 max=0.3;
    yaxis label="Hitting Efficiency";
    title "Scatter Plot of Defensive Efficiency vs Hitting Efficiency (Top 100 Players)";
run;
```

Players positioned further to the right are better defensively, while those positioned higher demonstrate better hitting efficiency.
From the graph, we can determine:

- **Best hitter:** Jesse Barfield
- **Best defender:** Darrell Evans
- **Best overall player:** Don Mattingly

4.3 Do More Experienced Players Perform Better Offensively?

To test whether there is a correlation between hitting efficiency and experience level we decided to conduct a Kruskal-Wallis test. This is a non-parametric test that will test for differences in the means for years of experience.

We can visualise the test in a Wilcoxon Scores Box Plot where each player's hitting efficiency is assigned a Wilcoxon score based on how it ranks within the dataset. These scores are then used in a Kruskal-Wallis test to compare the years of experience.

We use the following code to create the Wilcoxon Scores Box Plot:

```
proc npar1way data=GP.HITS wilcoxon;
    class "no of years in major league"n;
    var weighted_hitting_efficiency;
run;
```

Kruskal-Wallis Test for Hitting Efficiency Across Experience Levels
Hypotheses:

- **Null Hypothesis** (H_0): There is no significant difference in hitting efficiency across different years of experience in the major league.
- **Alternative Hypothesis** (H_1): At least one experience group has a significantly different hitting efficiency distribution.

Test Results:

- **Chi-Square Statistic**: 42.5237
- **p-value**: 0.0036

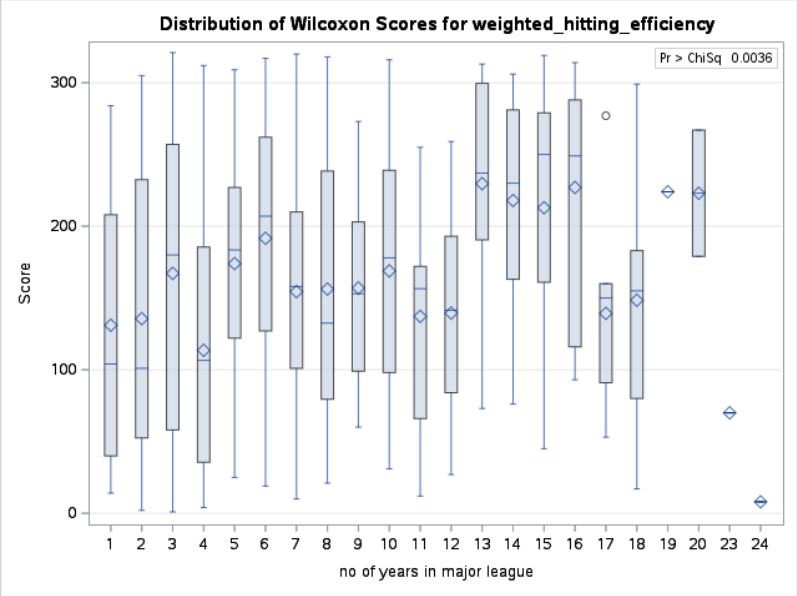


Figure 7: Distribution of Wilcoxon Scores for Hitting Efficiency by Years in Major League

Conclusion: Since the p-value ($p = 0.0036$) is less than the significance level ($\alpha = 0.05$), we **reject the null hypothesis**. This indicates that there is a statistically significant difference in hitting efficiency among players with different years of experience in the major league.

Although we can see there is difference in hitting with changes in experience we can't using the Kruskal-Wallis Test be certain that there is a positive correlation between the two. To test for a correlation we use the Spearman's correlation test as it is a non-parametric test.

Spearman's Correlation Test for Hitting Efficiency and Experience

To examine the relationship between weighted hitting efficiency and the number of years in the major league, we conducted a non-parametric Spearman's rank correlation test at a 5% significance level ($\alpha = 0.05$).
Hypotheses:

- **Null Hypothesis (H_0)**: There is no correlation between weighted hitting efficiency and the number of years in the major league ($\rho_s = 0$).
- **Alternative Hypothesis (H_1)**: There is a significant correlation ($\rho_s \neq 0$).

Test Results:

- **Spearman's correlation coefficient** (ρ_s): 0.16795
- **p-value**: 0.0025

Conclusion: Since the p-value ($p = 0.0025$) is less than the significance level ($\alpha = 0.05$), we **reject the null hypothesis**. This indicates a statistically significant but weak positive correlation between weighted hitting efficiency and years of experience in the major league.
A weak positive correlation shows that there is a slight increase in hitting efficiency for players that have more years of experience in the major league.

4.4 Do More Experienced Players Perform Better Defensively?

Similarly to the comparison between hitting and experience we can once again use a Kruskal-Wallis test. This will test for the difference in the means of the defensive efficiency for each year of experience.

We can visualise the test in a Wilcoxon Scores Box Plot, where each player's defensive efficiency is assigned a Wilcoxon score based on how it ranks within the dataset.

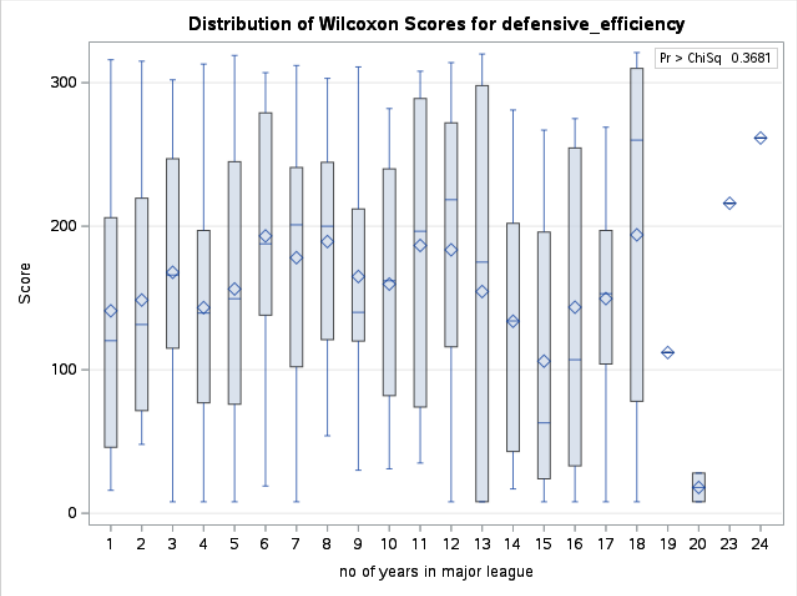


Figure 8: Distribution of Wilcoxon Scores for Defensive Efficiency by Years in Major League

Kruskal-Wallis Test for Defensive Efficiency Across Experience Levels
Hypotheses:

- **Null Hypothesis** (H_0): There is no significant difference in defensive efficiency across different years of experience in the major league.
- **Alternative Hypothesis** (H_1): At least one experience group has a significantly different defensive efficiency distribution.

Test Results:

- **Chi-Square Statistic**: 22.5564
- **p-value**: 0.3681

Conclusion: Since the p-value ($p = 0.3681$) is greater than the significance level ($\alpha = 0.05$), we **fail to reject the null hypothesis**.
This indicates that there is no statistically significant difference in defensive efficiency among players with different years of experience in the major league. Although we can observe changes in defensive efficiency across experience levels, the Kruskal-Wallis Test does not show a significant difference in defensive efficiency based on years of experience.

4.5 Does salary vary depending on team and league?

To compare the salaries of teams in the AL and the NL, we first had to calculate the teams average salary, as our raw data had only individual player's salaries. We then were able to plot this on a graph to clearly show the variation of salary across teams. To do so we used the following code:

```
proc sgplot data=SOR2002.hitsnew;
    vbar"team 86"n / response=salary_num group="lague 86"n stat=mean
              datalabel;
    xaxis label="Team";
    yaxis label="Average Salary";
    title "Average Salary of Teams by League";
run;
```

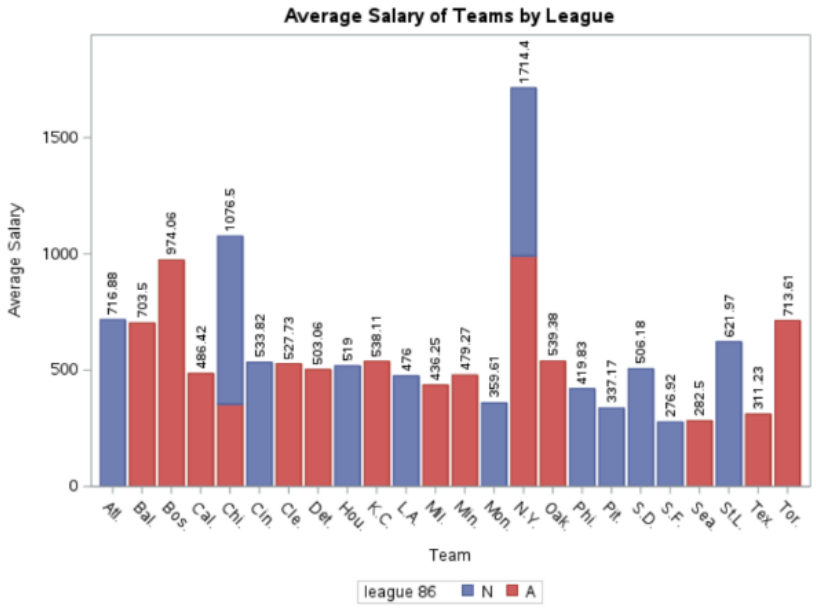


Figure 9: The average salaries of Teams that played in the AL and the NL

From Figure 4 we can see that teams in the AL receive a higher average salary, with Boston Red Sox having the highest average salary of $974,060. From the figure, also notice that Chicago and New York are being displayed as having a salary for the AL and the NL. This is because both these cities had two teams that played in major leagues. New York had the New York Mets that played in the AL and New York Yankees played in the NL, while Chicago had Chicago White Sox that played in the AL and Chicago Cubs that played in the NL.

Once we could visualise the variation in the average salaries across the teams in the two leagues, we wished to test the relationship of the two variables. To do so, we used the PROC NPAR1WAY procedure to perform a non-parametric test on our data.

```
proc npar1way data=SOR2002.hitsnew;
    class "no of years in major league"n;
    var salary_num;
run;
```

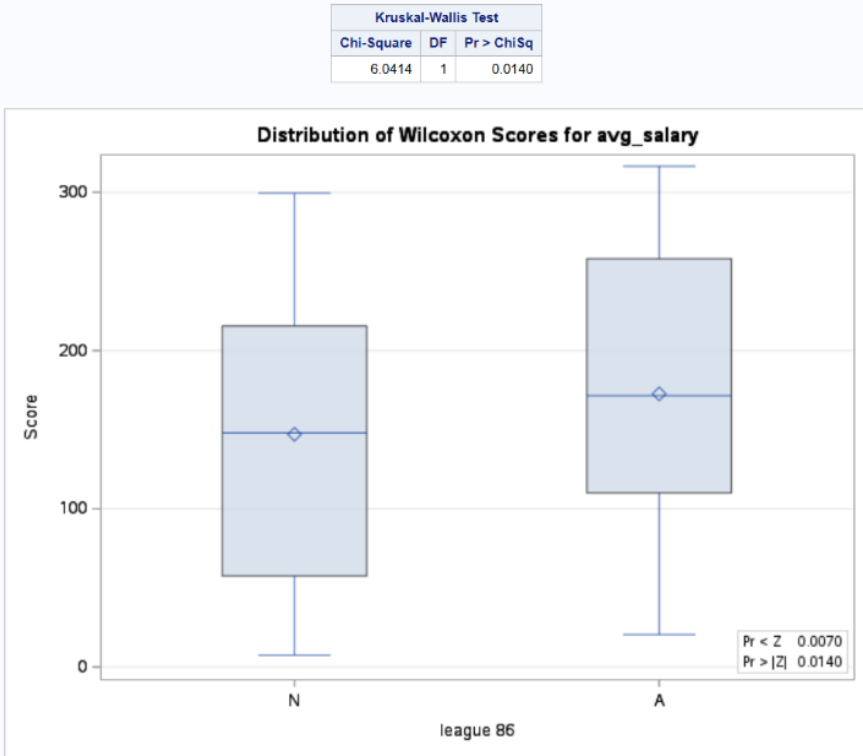


Figure 10: Kruskal-Wallis test and the distribution of Wilcoxon scores for average salary against league

Hypotheses

- **Null Hypothesis (H_0)**: There are no differences in the average salary distributions of the teams in the two leagues.
- **Alternative Hypothesis (H_1)**: There is a difference in the average salary distributions of the teams in the two leagues.

Test Results:

- **Chi-Square Statistic**: 6.0414
- **p-value**: 0.0140

Conclusion: Since the p-value ($p = 0.0140$) is less than the significance level ($\alpha = 0.05$), we can **reject the null hypothesis** in favour of **the alternative hypothesis**. That is, there is a difference in the average salary distributions, meaning the average salary of the teams is dependent on the league they are in.

4.6 Do more experienced players earn a higher salary?

To investigate whether the player's salary is dependent on the number of years they have played in the major league, we can perform the PROC NPAR1WAY procedure again. Using a similar code, we can produce another Kruskal-Wallis test.

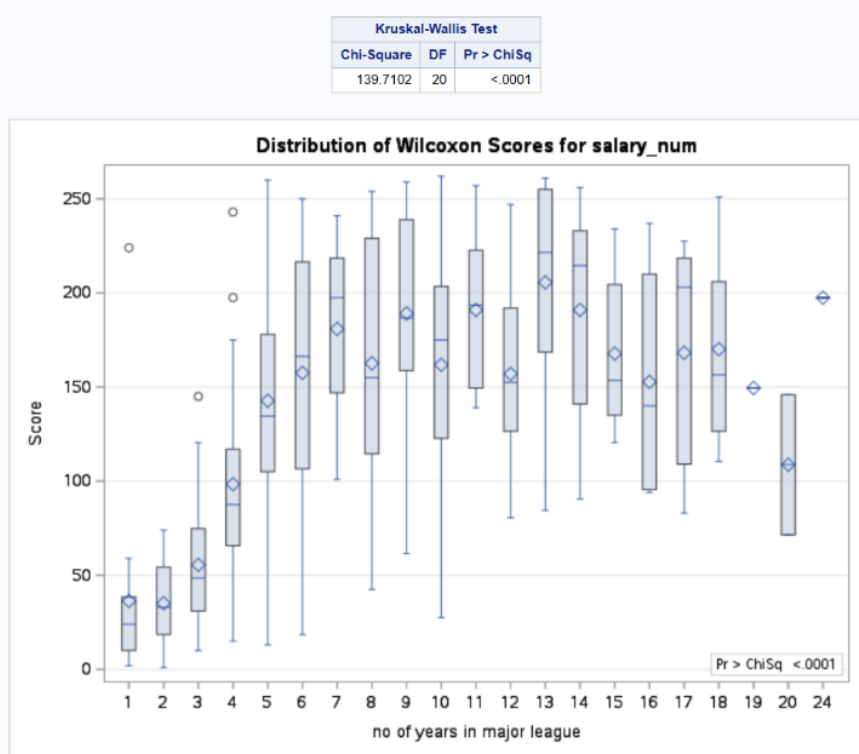


Figure 11: Kruskal-Wallis test and the distribution of Wilcoxon scores for number of years in the Major League against salary

Hypotheses

- **Null Hypothesis (H_0)**: The number of years a player has played in the Major League does not affect their salary.
- **Alternative Hypothesis (H_1)**: The number of years a player has played in the Major League does affect their salary.

Test Results:

- **Chi-Square Statistic**: 139.7102
- **p-value**: <0.0001

Conclusion: Since the p-value ($p < 0.0001$) is less than the significance level ($\alpha = 0.05$), we can **reject the null hypothesis** in favour of **the alternative hypothesis**. That is, the number of years a player has played in the major league does affect their salary. This suggests that more experienced players earn a higher salary. However, from Figure 6 we can see salaries of players who have played in the Major League for more than 13 years begin to decline, which could be caused by performance decline, the ageing curve and numerous other factors.

4.7 Modeling salaries of Major league Baseball Players

Using linear regression, we can consider modelling the player's salaries for the 1987 season based on batting statistics from the 1986 season and whole career batting performance. Suppose the

proposed model E($\mathbf{Y}$) is a linear function of known functions, where $\mathbf{Y}$ is the salary variable we wish to model, we use the model

$$E(\mathbf{Y}) = \mathbf{X}\beta$$

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$$

assuming,

$$\text{Var}(\mathbf{Y}) = \text{Var}(\mathbf{e}) = \sigma^2\mathbf{I}_n$$

where the variances are equal and the covariance equal 0. We've acknowledged that there is high levels of variation for higher salaries, so we can apply a log transformation for this analysis. To do so, we use the following code:

```
ods graphics on;

proc reg data=SOR2002.hitsnew;
    id name "team 86"n "league 86"n;
    model logSalary = "no of hits 86"n
                      "no of runs 86"n
                      "no of runs batted 86"n
                      "no of walks 86"n
                      "no of years in major league"n
                      "no of hits during career"n;
run;
```

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	113.78291	18.96382	53.35	<.0001
Error	254	90.29071	0.35548		
Corrected Total	260	204.07362			

Root MSE	0.59622	R-Square	0.5576
Dependent Mean	5.91888	Adj R-Sq	0.5471
Coeff Var	10.07315		

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	4.34377	0.14103	30.80	<.0001
no of hits 86	no of hits 86	1	0.00530	0.00218	2.43	0.0156
no of runs 86	no of runs 86	1	0.00068481	0.00413	0.17	0.8683
no_of_runs_batted_86	No of runs batted 86	1	0.00102	0.00245	0.42	0.6758
no of walks 86	no of walks 86	1	0.00608	0.00247	2.46	0.0144
no of years in major league	no of years in major league	1	0.04822	0.02086	2.31	0.0216
no of hits during career	no of hits during career	1	0.00043286	0.00016636	2.60	0.0098

Figure 12: Default output produced by the PROC REG procedure

From Figure 12 we can see the P-values of both "no of runs 86" and "no of runs batted 86" are both high. This indicates that these two variables do not highly affect the model for predicting salary, so we can omit these variables from our model.

Once we omit these two variables we can obtain Fit Diagnostics to examine the influences on logsalary.

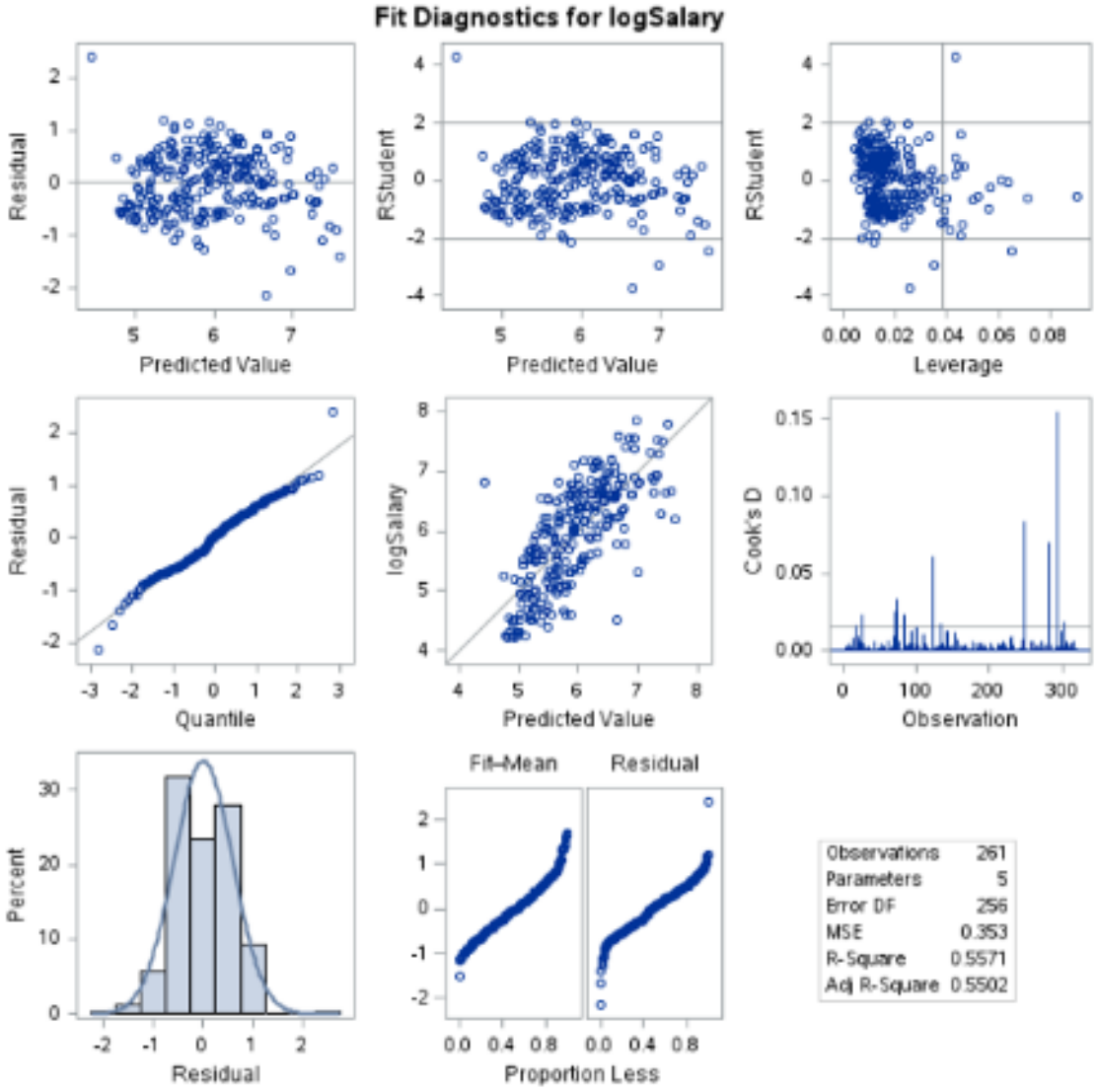


Figure 13: Fit Diagnostics for logsalary

We can see that there are still several influential and outlying observations. One possible reason for observing outliers is that the linear model specified is not appropriate to capture the variation in this data. We can often see evidence of an inappropriate model by observing patterns in plots of residuals.



Figure 14: Plots of the Residuals by the regressors in the model

From above we can see that the players near the start of their careers and players near the end of their careers get paid less than the model predicts. To address this lack of fit by using polynomials of degree 2 for these two variables as shown in the following statements:

```
proc reg data=SOR2002.hits3
      plots=(diagnostics(stats=none) RStudentByLeverage(label)
             CooksD(label) Residuals(smooth)
             DFFITS(label) DFBETAS ObservedByPredicted(label));
   id name "team 86"n "league 86"n;
   model logSalary = "no of hits 86"n
                     "no of walks 86"n
                     "no of years in major league"n
                     "no of hits during career"n
                     YrMajor2
                     CrHits2;
run;
```

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	151.66220	25.27703	122.50	<.0001
Error	254	52.41142	0.20634		
Corrected Total	260	204.07362			

Root MSE	0.45425	R-Square	0.7432
Dependent Mean	5.91888	Adj R-Sq	0.7371
Coeff Var	7.67461		

Parameter Estimates

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	3.94906	0.12326	32.04	<.0001
no of hits 86	no of hits 86	1	0.00188	0.00094596	1.99	0.0476
no of walks 86	no of walks 86	1	0.00596	0.00162	3.67	0.0003
no of years in major league	no of years in major league	1	0.21875	0.03704	5.91	<.0001
no of hits during career	no of hits during career	1	0.00174	0.00029731	5.86	<.0001
YrMajor2		1	-0.01375	0.00179	-7.69	<.0001
CrHits2		1	-3.20832E-7	1.080271E-7	-2.97	0.0033

Figure 15: The analysis of variance and parameter estimates for this model

From the figure above, notice that we now have a larger R-square value of 0.7432 which is significantly larger than the R-square value we obtained from Figure 12 (0.5676). This indicates that the predictors we have used in this model are better than the predictors we have used beforehand to capture the underlying patterns in the data. Although this larger R-square value can be a sign of a better fitting model, it is not always the ultimate indicator of a good model.

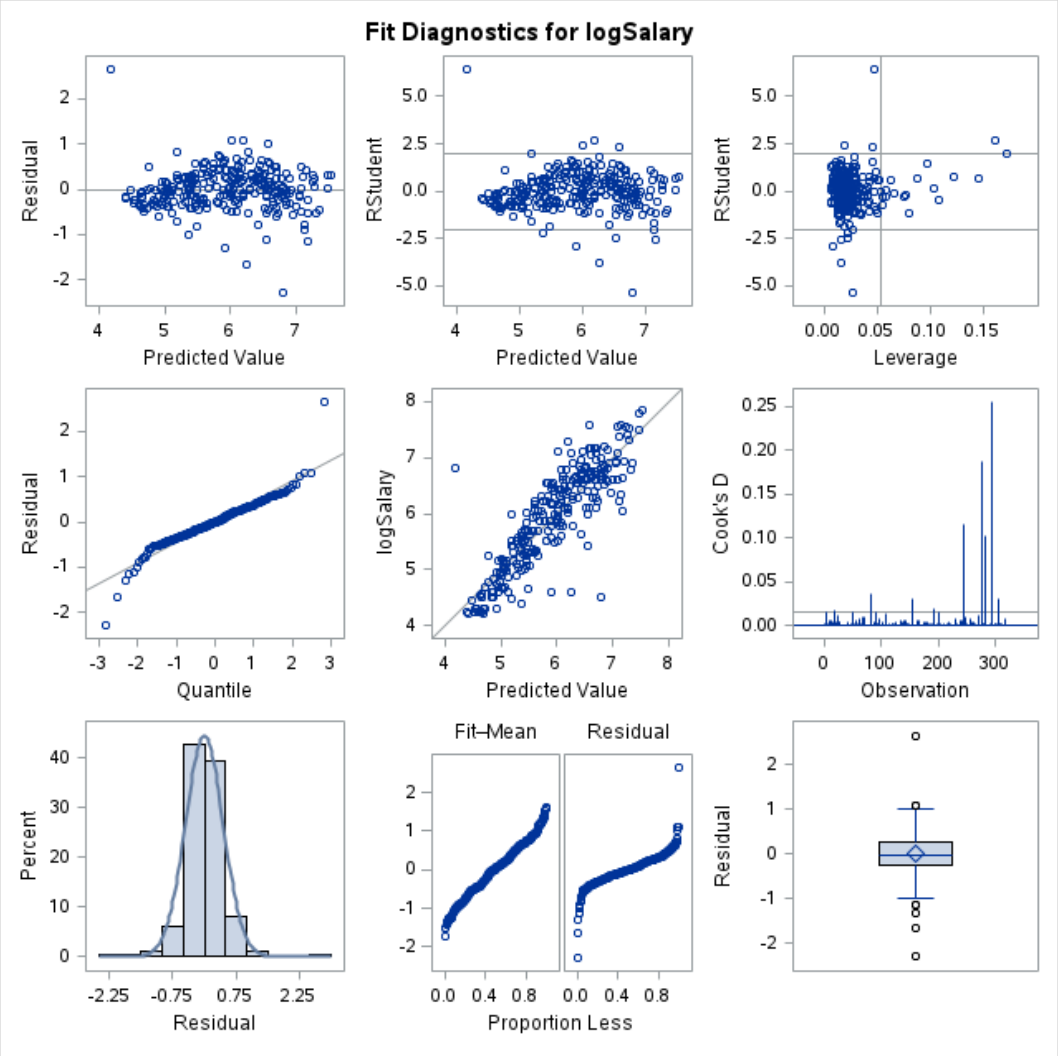


Figure 16: The distribution of the residuals that can be seen in a histogram, Q-Q plot, and box plot

The observed by predicted value plot, from the distribution of residuals reveals a reasonably successful model for explaining the variation in salary for most of the players. However, the model tends to over-predict the salaries of several players near the lower end of the salary range. [SAS]

The over prediction of salaries may have been a result of the exploitation of some players. An Economist Andrew Zimbalist found that the players in 1986 were not being paid to reflect their true value to their team. However, from 1986 to 1989 he had seen the average exploitation of players decrease from 1.32 to 1.01. He found that younger players had the highest degree of exploitation, as they had little bargaining power,then the level fell with players in the 2-5 year category and eventually disappeared completely for players with 6 or more years experience. Zimbalist had found that the most experienced players were often overpaid. There was also no similar studies completed before free agency, so our model is based on players who may have been substantially overpaid, such as the veteran players. In turn our model may have incorrectly assumed that the salary inflation for veteran players would continue at the same rate for all players. [Exploitation patterns]

5 Conclusions

Based on the analyses presented, several notable insights can be observed from the Hits dataset:

5.1 No Meaningful Correlation Between Hitting and Defence

Both scatter plots (17, 6) and a Spearman's correlation test revealed no significant relationship between offensive efficiency (e.g., hits, runs, runs batted in) and defensive efficiency (e.g., putouts, assists, errors). Even when filtering out outliers or focusing on the top 100 most frequent batters, the results remained the same.

This can be intuited when thinking about how different these 2 different aspects of the game are played and the athletic qualities required. Hitting a baseball hard and accurately is very different from positioning yourself correctly on the field to catch and try to put out your opponent, so it will not have a strong correlation, a small player would be disadvantaged on defense due to less reach or power for jumps, while a tall player may be slower at hitting. This can be seen in other sports like basketball where some of the best offensive forces in the game are notoriously bad defenders, or in football were defenders pretty much never score.

By analysing the top 100 most frequent batters in Figure 3, Jesse Barfield emerged as the best hitter and Darrell Evans as the best defender. When both offensive and defensive efficiency were considered, Mattingly emerged as the best overall player in the 1986 season. Although these players are great at offence and defence and do not clearly follow our conclusion, they are relative outliers to the general trend and do not change the statistical findings when included in the data, this is to be expected as the very best players of any sport will need to be good at all aspects of the game.

5.2 Experience Impacts Hitting Efficiency—But Only Slightly

A Kruskal-Wallis test showed that players with different years of Major League experience had statistically different distributions of hitting efficiency.

Spearman's correlation confirmed a weak (but significant) positive correlation between experience and hitting performance. In practical terms, this suggests that hitters gain small benefits from experience in the league every year and that as the years go by, these gains add to each other, leading to a bigger and bigger gap in hitting efficiency, as seen in Figure 1. This makes sense as hitting is a repetitive simple skill that can be practiced 100s of times a day, which allows players to perfect every little nuance of its action allowing for consistent increase in efficiency.

5.3 No Significant Difference in Defensive Efficiency by Experience

In contrast to hitting efficiency, a similar Kruskal-Wallis test for defensive efficiency across experience levels did not reveal significant differences. Therefore, having many years in the league does not necessarily guarantee stronger defensive performance.

This makes sense as defending has a far greater number of unpredictable variables that go into it, knowing were to stand to catch the ball, for example, is a very intuitive skill that is much harder to get good at if you don't already have the instincts for it. The reach of a player will not increase with time to allow better catching as opposed to the muscles of the player getting stronger to hit.

5.4 Salary Varies Significantly by League and Team

A Kruskal–Wallis test confirmed a statistically significant difference in salary distributions between the two leagues as seen in Figure 10 , meaning salary levels are at least partly dependent on whether a team is in the AL or NL. When comparing the average salaries of teams in the American League (AL) and National League (NL), AL teams tended to have a higher overall salary. This implies that American league teams have more ticket sales and merchandising to fund their players inflated salaries and this is supported by the fact that several of the most iconic franchises (e.g., the New York Yankees and the Boston Red Sox) play in the AL, attracting more fans, higher ticket sales, more lucrative sponsorships, and larger TV deals which will pull the league average salary up significantly.

However, we can also see from Figure 9 that salary also varies significantly by team even in the same league. The Seattle Mariners earn less than half the annual salary of the Boston Red Sox so it's not just the leagues that show a wage disparity. This can be explained by geographic factors such as location and weather. More populated areas with better weather such as New York will have higher paid teams than those in more desolate cold areas. This could be used by potential owners to decide were to start potential new franchises.

5.5 More Experienced Players Generally Earn Higher Salaries

Another Kruskal-Wallis test and a distribution graph of Wilcoxon scores for number of years in the Major League against salary (11) showed a strong relationship between years played in the league and salary. As players accumulate more time in a league, their salaries tend to rise—at least until very high levels of experience (around 13+ years), where salaries appear to plateau or slightly decline. This is reinforced by the fact that offensive efficiency increases with time and so a players value will increase consistently with experience, but will eventually be outpaced by the decline due to age and injury accumulation.

5.6 Modelling Salary of Major League Baseball players

We initially used multiple linear regression model with a log-transformed salary due to the large variation in high end salaries. It initially left some large residuals and did not fully capture salary variation. We removed 'no of runs 86' and 'no of runs batted 86' as they did not highly effect the model for predicting salary. Using fit diagnostics revealed our model still had several outlying observations and that our model tended to overvalue players at the start and end of their career.(13,14)

Adding polynomial terms for 'no of hits during career' and 'no of years in the major league' improved the model fit substantially (R^2 increased to 0.7432), indicating these predictors help account for non-linear relationships.(15)

Despite the improved model, a few extreme outliers remained specifically towards to lower end salaries(16), suggesting that additional factors (e.g., position scarcity, market size, or contract negotiation specifics) influence salary beyond standard performance metrics.

6 Further Analyses and Additional Data Considerations

6.1 Player Consistency vs Variability

An analysis of player performance variability could provide deeper insight into performance trends. If the dataset included quarterly or monthly statistics, it would be possible to track variations over the season using moving averages. This approach would help identify slumps and peak periods, allowing for the quantification of risk factors associated with playing individual players at specific times and allow managers to try and mitigate any negative findings. While this analysis would require additional time, the data could have been structured accordingly at the time of collection without much more effort but allow for much more detailed insights.

6.2 Injury and Fatigue Factors

Understanding the impact of injuries and workload on player performance is crucial. A historical injury record could help determine whether past injuries negatively affected a player's performance, particularly in cases where salaries appear disproportionately high compared to on-field output. This could explain apparent biases in contract decisions, such as highly paid yet underperforming players who were injured in the past (e.g., injured superstars).

Additionally, analysing the number of games played per season could reveal patterns in workload management and potential injury correlations. Overuse may lead to performance declines, and incorporating such data would allow for more accurate evaluations of player durability and career longevity.

6.3 Player Demographics

Several demographic factors could influence player performance and career progression:

- Age, height, weight, and handedness could be analysed to determine their relationship with various performance metrics. This could help identify:
  - The age at which players reach peak performance and when they tend to decline.
  - The optimal height-to-weight ratio for different playing positions.
  - Whether left-handed or right-handed players demonstrate distinct advantages in specific aspects of the game.
- Birthplace or nationality could be examined to determine if geographic factors contribute to player success. This analysis could highlight potential disparities in player development, such as underrepresentation of foreign players due to differences in training infrastructure, scouting opportunities, or cultural emphasis on the sport.

Incorporating these additional variables would enhance the depth of our statistical analysis, providing a more comprehensive understanding of player performance dynamics.

References

[Burns] Ken Burns *Baseball for Beginners* https://www.pbs.org/kenburns/baseball/baseball-for-beginners (accessed 12/03/2025)

[MLB] Major League Baseball *Walk (BB)* https://www.mlb.com/glossary/standard-stats/walk?msockid=1975da8e2a3d6e9312bac82e2bdd6f5f (accessed 12/03/2025)

[Butler] Jason Butler (28/02/2025) *Baseball Abbreviations Explained* https://honestbaseball.com/baseball-abbreviations-explained/ (accessed 12/03/2025)

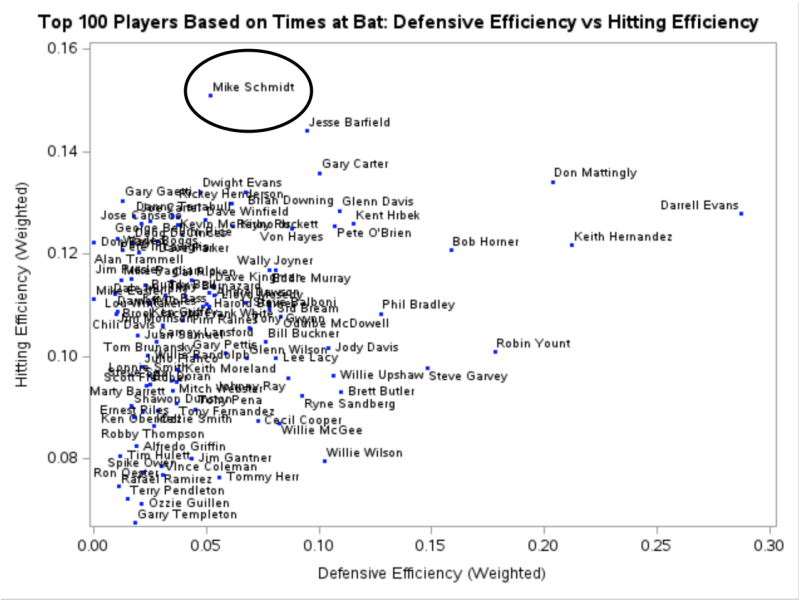
[SAS] SAS *Modeling Salaries of Major League Baseball Players* https://documentation.sas.com/doc/en/statug/latest/statug_reg_examples01.htm (accessed 05/03/2025)

[Baseball Almanac] *Mike Schmidt's 1986 Hitting Log* https://www.baseball-almanac.com/players/hittinglogs.php?p=schmimi01&y=1986 (accessed 05/03/2025)

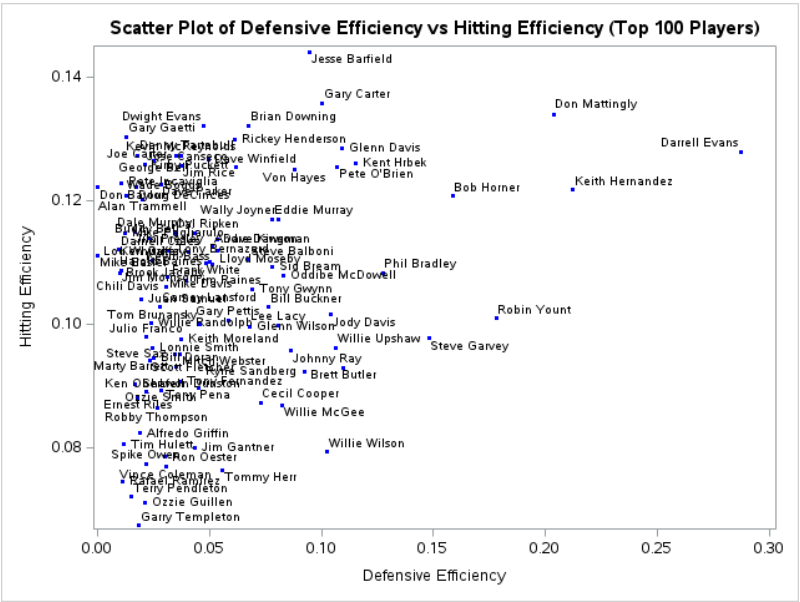
[Exploitation patterns] Michael J.Haupert *The Economic History of Major League Baseball* https://eh.net/encyclopedia/the-economic-history-of-major-league-baseball/ (accessed 19/03/25)

7 Appendix

We previously discussed how Mike Schmidt was incorrectly represented in our dataset. As the actual MVP (Most Valuable Player) of the 1986 season, we thought it would be interesting to see how his true stats would fit into our results.To do this, we researched his stats from the 1986 season and made a new dataset with his actual stats. We can compare the top 100 players graph previously seen with Mike Schmidt now included



(a) Dataset with Mike Schmidt



(b) Dataset without Mike Schmidt.

Figure 17: Scatter plots showing the relationship between hitting efficiency and defensive efficiency.

As we can see, Mike Schmidt was clearly the best offensive player in the 1986 season, and his inclusion emphasizes that MVP voting tends to favour hitting over defence.